

**DELHI PUBLIC SCHOOL**  
**ANNUAL EXAMINATION 2026 — SCIENCE**

Class X | Time: 3 Hours | Maximum Marks: 80

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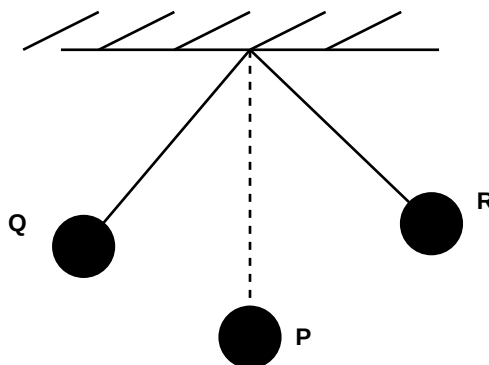
**General Instructions:**

1. This paper contains three sections: Section A, Section B and Section C.
2. All questions are compulsory unless a choice is indicated.
3. Marks for each question are indicated against it, where printed.
4. Draw diagrams wherever necessary and label them clearly.

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**SECTION A — PHYSICS**

1. Define the term 'inertia'. [2 marks]
2. State the SI unit of force and define one newton. [2 marks]
3. Solve for x:  $2x^2 - 5x + 3 = 0$ . Show all steps of your working. [5 marks]
4. The diagram below shows a simple pendulum released from rest at point R and allowed to swing through its lowest point P. Label points P, Q and R, and explain the energy transformation of the bob as it moves from R to P. [5 marks]



*Answer either Question 5 or Question 6.*

5. State and explain Newton's second law of motion, with one real-life example. [5 marks]
6. Derive the expression for the kinetic energy of a body of mass  $m$  moving with velocity  $v$ . [5 marks]
7. Why does ice float on water? Explain in terms of density.
8. Specific heat and heating of water.  
(a) Define specific heat capacity. [2 marks]

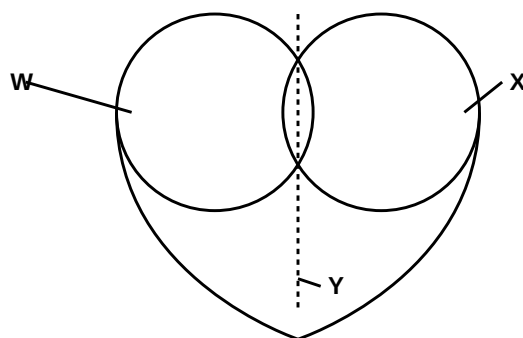
*(Section A continues on the next page)*

- (b) Calculate the heat required to raise the temperature of 2 kg of water from 20 degrees Celsius to 80 degrees Celsius. (Specific heat capacity of water = 4200 J/kg-degC) [3 marks]

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## SECTION B — BIOLOGY & CHEMISTRY

1. Name the powerhouse of the cell. [1 marks]
2. What is photosynthesis? Write the balanced chemical equation for photosynthesis. [4 marks]
9. List two differences between plant cells and animal cells. [4 marks]
10. Balance the following chemical equation:  $\text{Fe} + \text{O}_2 \rightarrow \text{Fe}_2\text{O}_3$  [3 marks]
11. Acids, bases and ionic conduction.
  - (a i) What is the pH of a neutral solution? [1 marks]
  - (a ii) Name one acid found in the human stomach. [1 marks]
  - (b) Explain why sodium chloride conducts electricity in its molten state but not in its solid state. [3 marks]
12. Draw a labelled diagram of the human digestive system. [5 marks]
13. The diagram below shows a simplified schematic of the human heart. Identify the parts marked W, X and Y. [5 marks]



## SECTION C — MATHEMATICS

14. Solve for  $x$  using the quadratic formula:  $x^2 - 7x + 10 = 0$ . Show all steps. [4 marks]
15. A ladder leans against a wall making an angle of 60 degrees with the ground. If the foot of the ladder is 5 m from the wall, find the length of the ladder. (Draw and label a rough diagram as part of your working.) [4 marks]
16. Find the volume and total surface area of a cylinder of radius 7 cm and height 20 cm. (Use  $\pi = 22/7$ ) [4 marks]

*Answer either Question 17 or Question 18.*

17. Prove that the square root of 2 is an irrational number. [3 marks]
18. Find the HCF and LCM of 96 and 404 using the prime factorisation method. [3 marks]